- Lipids protect the internal organs, serve as insulating materials and give shape and smooth appearance to the body.
- Fatty acids are the simplest form of lipids and they serve as the constituent in a large number of complex form of lipids.
- Biosynthesis of fatty acids in animals occurs mainly in the cytosol of adipose tissue, mammary glands, and liver. In case of plants, it occurs in the stroma of chloroplast.
- Palmitic acid is a 16-carbon saturated fatty acid. A sequence of 7 enzyme catalyzed reactions converts 2 carbons unit to 4 carbons unit. First three steps of biosynthesis involve formation of acetyl ACP and malonyl ACP from acetyl CoA.
- The digestion of fats starts in the small intestine. Fats are emulsified by bile salts and hydrolyzed by pancreatic lipases to form free fatty acids.
- The β-oxidation of fatty acids involves three stages namely activation of fatty acids, transport of activated fatty acids to mitochondria followed and the oxidation of the activated fatty acids.
- Fatty acids are oxidized by most of the tissues in the body. However, brain, erythrocytes and adrenal medulla cannot utilize fatty acids for energy requirement.
- Cholesterol is widely distributed in all cells and is a major component of the cell membrane and lipoproteins. Cholesterol is found exclusively in animals; hence it is often called as animal sterol.
- Cholesterol, when oxidized under suitable conditions, undergoes rapid oxidation to form a ketone called cholestenone.
- In the cholesterol biosynthetic pathway, the HMG-CoA is reduced to form mevalonate by NADPH dependent reductase (HMG-CoA reductase). This is the rate limiting enzyme in the pathway of cholesterol biosynthesis.
- Cholesterol is a precursor for other important steroid molecules such as the bile salts, steroid hormones, and vitamin D etc..
- Phospholipids are a characteristic constituent of cellular membranes. A phospholipid is an amphipathic molecule constructed from four components: fatty acids, a platform to which the fatty acids are attached, a phosphate and an alcohol attached to the phosphate. The platform on which phospholipids are built may be glycerol or sphingosine.
- Phospholipids can be synthesized by two mechanisms. One utilizes a CDP-activated polar head group for attachment to the phosphate of phosphatidic acid. The other utilizes CDP-activated 1,2-diacylglycerol and an inactivated polar head group.

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